



GLOBAL
BIOTECH

글로벌바이오테크

LIFE SCIENCE SOLUTIONS

— E N D O T O X I N A N A L Y S I S —

Endotoxin Analysis and New Software

based on

MRX MICROPLATE READER

K LAB introduces the future development plans and capabilities of endotoxin analysis using the MRX microplate reader. With innovative technology, we are opening a new chapter in life science research.

C O N T A C T

글로벌바이오테크

T. 02-406-4387

정기운 팀장 010-4253-4387



— M R X S E R I E S —

1. Molecular Spectroscopy



POP

Single-Beam
UV-Vis Spectrophotometer



QX

Water Analysis
Spectrophotometer



Alpha

High-Performance Double-Beam
UV-Vis Spectrophotometer

2. Life Sciences Spectrophotometry



NanoQ

PDA Type · Full Spectrum
Microvolume Spectrophotometer



NanoQ Lite

LED Type · Compact
Microvolume Spectrophotometer



MRX

Microplate Reader
UV-Vis Spectrophotometer

02

Endotoxin Overview

Overview of endotoxin analysis methods and procedures

Definition and Risk of Endotoxin

What is Endotoxin?

Endotoxin is a lipopolysaccharide (LPS) found in the cell walls of gram-negative bacteria, and is released when bacteria die or break down.

Effects on the Human Body

If endotoxins enter the bloodstream, they can cause inflammatory reactions due to immune response — potentially leading to organ failure, septic shock, and even death.

1. ENDOTOXIN OVERVIEW

How do Endotoxins Enter the Body?

Medical Product Contamination

Injectables

Medical Devices

Pharmaceuticals

Why Endotoxin Testing is Essential

0 1

**Prevent
Contamination**

0 2

**Enhance
Product Quality**

0 3

**Protect
Consumers**

2. ENDOTOXIN METHOD

Types of Endotoxin Analysis Methods

METHOD 01

Optical Method

Measures changes in absorbance or turbidity to detect endotoxins.
Objective, quantitative, and well-suited for automation.

METHOD 02

Gel Clot Method

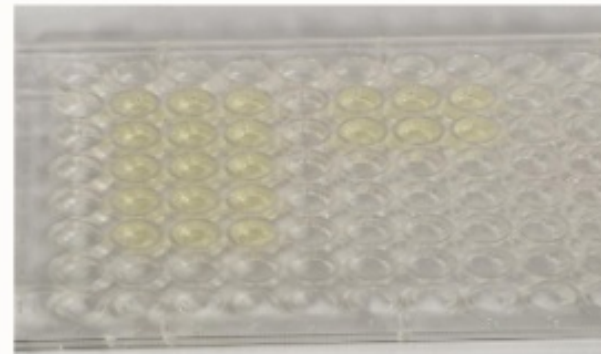
A traditional method where the formation of a gel indicates the presence of endotoxins. Simple, but subjective.

Types of Optical Methods

1. Chromogenic

The chemical reaction between endotoxins and LAL reagent causes a colour change, which is measured to determine concentration.

Change to yellow colour



2. Turbidimetric

Endotoxins react with the LAL reagent to form clots, increasing turbidity. This turbidity is measured to detect endotoxins.

Turbidity formation



3. M R X K E Y F E A T U R E

Endotoxin Analysis Capabilities

01 Precise Temperature Control

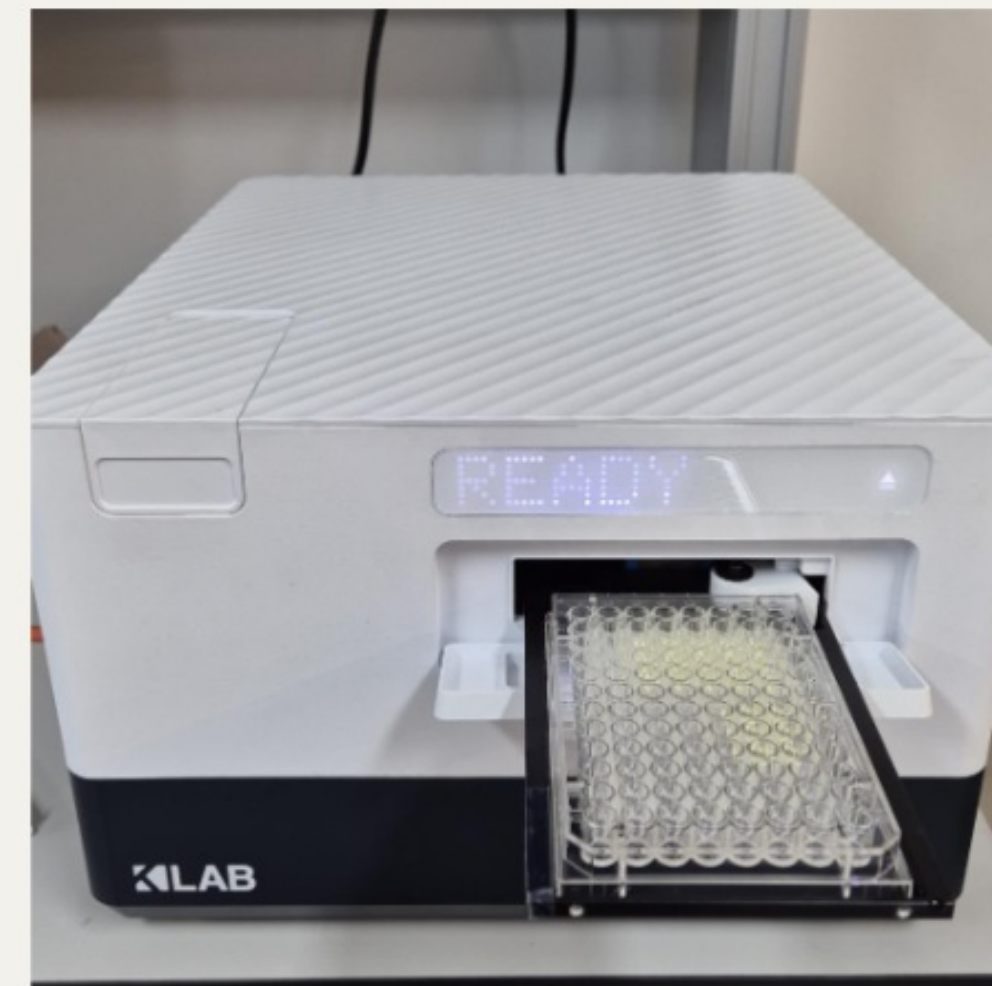
Incubating function controls analysis at 37 °C with ± 0.5 °C accuracy, ensuring optimal experimental conditions.

02 High-Speed Analysis

Reads a 96-well plate in under 8 seconds, enabling efficient processing of large sample volumes.

03 Real-Time Data Monitoring

Measured data can be checked in real time via software, with reports generated from live results.



03

Procedure of Endotoxin Analysis

1. EXPERIMENT PURPOSE

Purpose of Endotoxin Analysis with MRX

**Validate MRX for reliable
endotoxin analysis.**

1. EXPERIMENT PURPOSE

Validation Metrics

M E T R I C 0 1

RSQ Value

≥ 0.98

M E T R I C 0 2

Negative Control Separation

$> 15 \%$

M E T R I C 0 3

PPC Recovery

50 ~ 200 %

Endotoxin Sample Preparation

①

Negative Control

LAL Water 100 $\mu\ell$

LAL Reagent 100 $\mu\ell$

②

Standard Solution

0.005 ~ 5 EU/mL

Standard 100 $\mu\ell$

LAL Reagent 100 $\mu\ell$

③

Product Sample

Sample 100 $\mu\ell$

LAL Reagent 100 $\mu\ell$

④

Positive Control

Sample 100 $\mu\ell$

LAL Reagent 100 $\mu\ell$

Spike 0.05 EU/mL 10 $\mu\ell$

3. M E A S U R E M E N T

Endotoxin Measurement Conditions

Chromogenic Method



Turbidimetric Method



3. M E A S U R E M E N T

Measurement Setup



Select Mode

Absorbance

Normal

Endpoint

Kinetic

Spectrum

Well Scan

Enzyme Kinetics

Nucleic Acid

dsDNA

RNA

ssDNA

Protein

BSA

IgG

Lysozyme

Cancel

3. M E A S U R E M E N T

Measurement Conditions

01 D U R A T I O N

Total 2 hours

Interval 30 seconds

Measurement Settings Kinetic

Wavelengths	Total	2	:	0	:	0
Plate Type	Interval	0	:	0	:	30
Read Area	Repeat Count	240				
Timing	Measurement Minimum Time	00:00:08				
Shake	<button>Apply Minimum Time</button>					
Others						

Default Ok Cancel

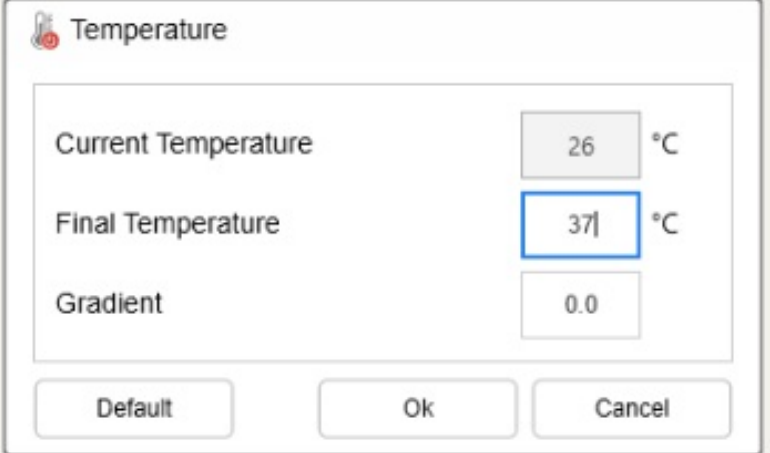
3. M E A S U R E M E N T

Measurement Conditions

0 2 T E M P E R A T U R E

37 °C

maintained during the 2-hour measurement

A dialog box titled "Temperature" with a thermometer icon. It contains three input fields: "Current Temperature" with the value "26", "Final Temperature" with the value "37" (highlighted with a blue border), and "Gradient" with the value "0.0". Each field is followed by a "°C" unit. At the bottom are three buttons: "Default", "Ok", and "Cancel".

Parameter	Value	Unit
Current Temperature	26	°C
Final Temperature	37	°C
Gradient	0.0	

3. MEASUREMENT

Wavelength

Chromogenic **405 nm**

Turbidity **340 nm**

Measurement Settings ⌕ Kinetic

Wavelengths

Plate Type

Read Area

Timing

Shake

Others

λ nm Add

340 ×

Default Ok Cancel

Measurement Settings ⌕ Kinetic

Wavelengths

Plate Type

Read Area

Timing

Shake

Others

λ nm Add

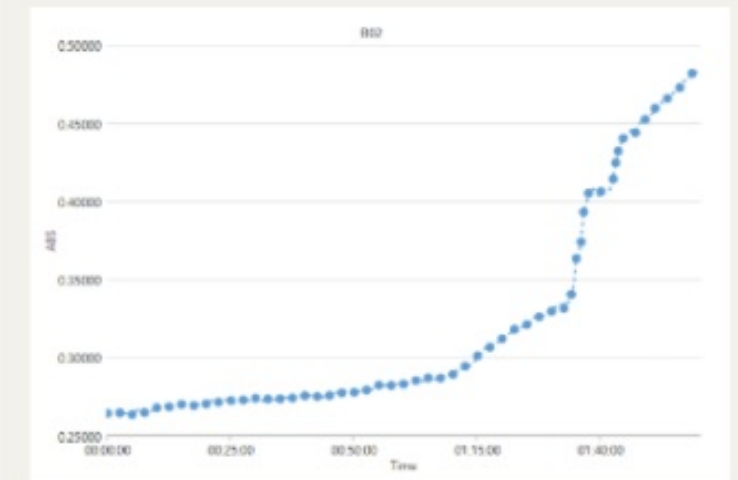
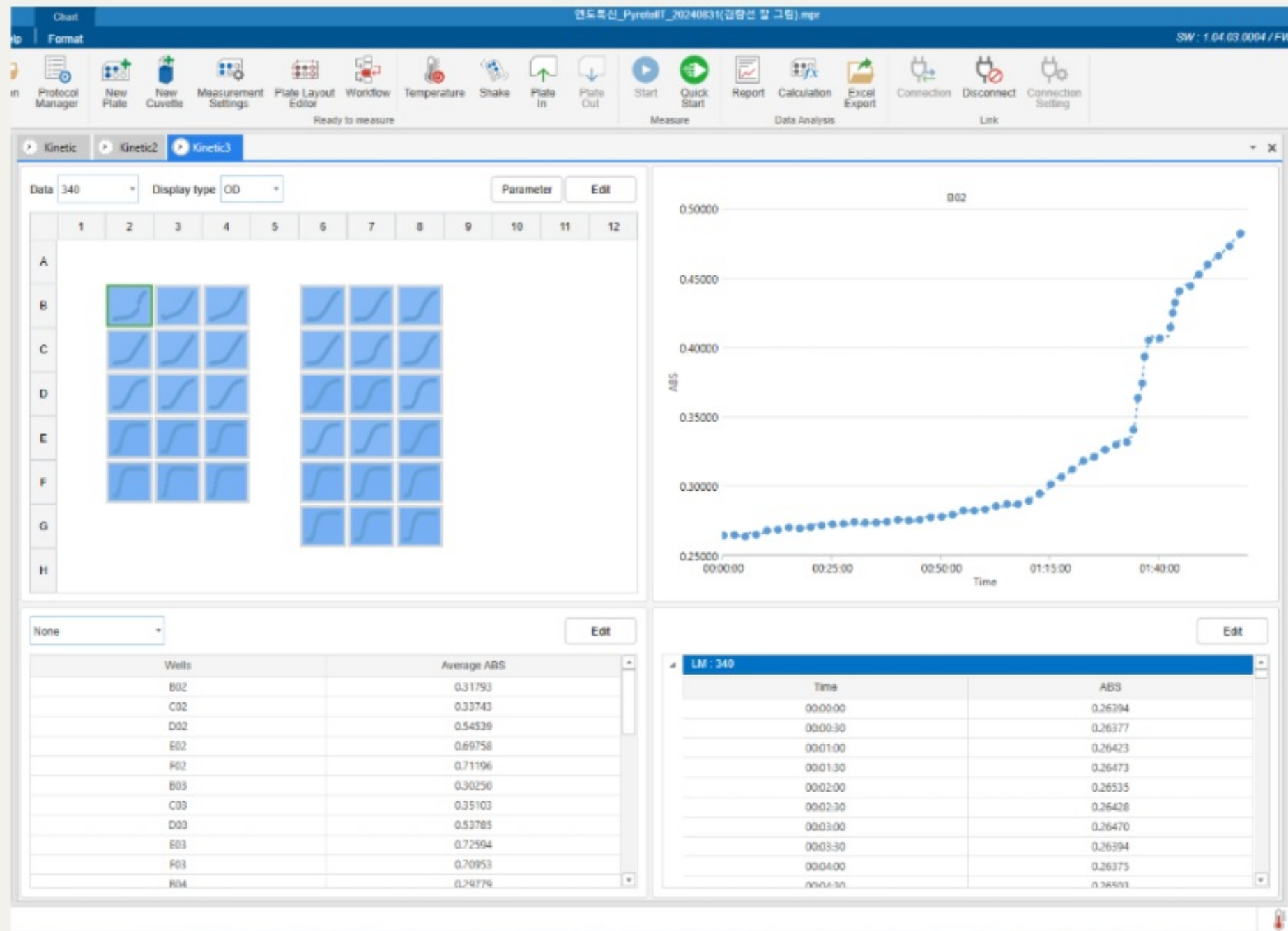
405 ×

Default Ok Cancel

4. DATA PROCESSING

Kinetic Mode Results

Kinetic mode outputs time-based OD values.



4. DATA PROCESSING

Determine Onset Time

01 OD Zero Calculation

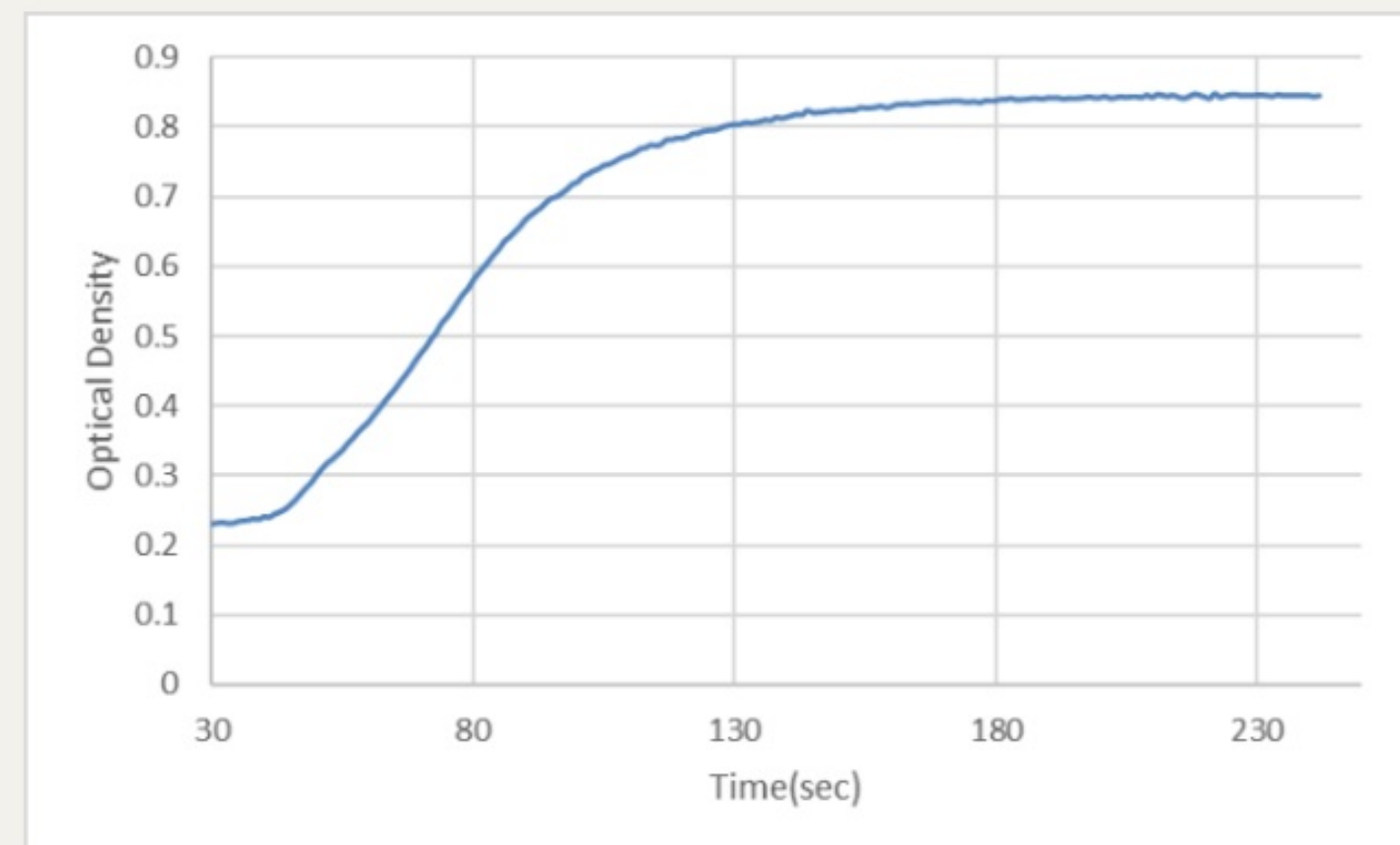
Average OD between 30 – 120 seconds.

02 Threshold OD Calculation

OD Zero + 0.03 (Chromogenic) / + 0.1 (Turbidimetric)

03 Onset Time Calculation

Time when sample OD reaches the threshold.



04

Results of Endotoxin Analysis

Sharing results of endotoxin analysis using MRX

RSQ Value Validation Overview

PURPOSE

Validate standard curve accuracy

METRIC

RSQ $\geq$ 0.98

METHODS

Chromogenic & Turbidity

VALIDATION — 01

RSQ Value Results

Turbidity Method

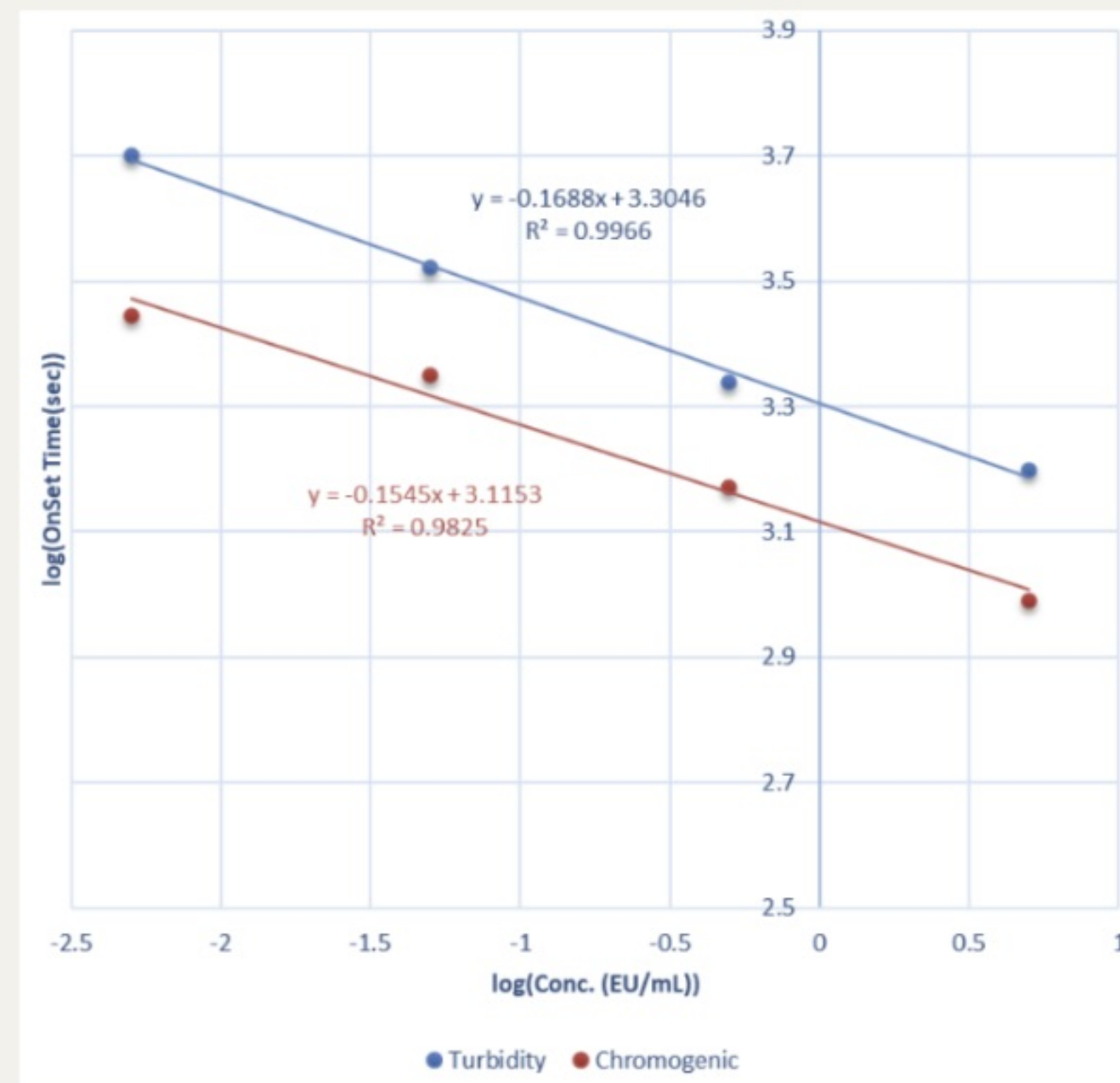
$RSQ \geq 0.9966$

Chromogenic Method

$RSQ \geq 0.9825$

Both $RSQ \geq 0.98$

BOTH PASS



Negative Control Separation Overview

PURPOSE

Ensure separation between negative control and detection limit

METRIC

> 15 %

METHODS

Chromogenic & Turbidity

Negative Control Separation Results

Chromogenic

Sample Name	Separation (%)	Determination
Negative Control-1	27.53	PASS
Negative Control-2	26.48	PASS
Negative Control-3	25.95	PASS

Turbidity

Sample Name	Separation (%)	Determination
Negative Control-1	28.33	PASS
Negative Control-2	28.33	PASS
Negative Control-3	27.39	PASS

All tests have received a PASS determination.

PPC Recovery Rate Overview

PURPOSE

Verify PPC recovery rate for accurate endotoxin detection

METRIC

50 % ~ 200 %

METHODS

Chromogenic & Turbidity

PPC Recovery Rate Results

Chromogenic

Sample Name	Recovery (%)	Determination
Positive Control-1	94.1	PASS
Positive Control-2	92.9	PASS
Positive Control-3	93.5	PASS

Turbidity

Sample Name	Recovery (%)	Determination
Positive Control-1	81.4	PASS
Positive Control-2	85.6	PASS
Positive Control-3	81.1	PASS

All tests have received a PASS determination.

Validation Conclusion

0 1

Validation Success

All metrics — RSQ, Negative Control, PPC — passed the required standards.

0 2

MRX Feasibility

MRX has been successfully validated for endotoxin analysis.

0 3

Next Steps

Focus on developing enhanced endotoxin analysis functionality for MRX.

05

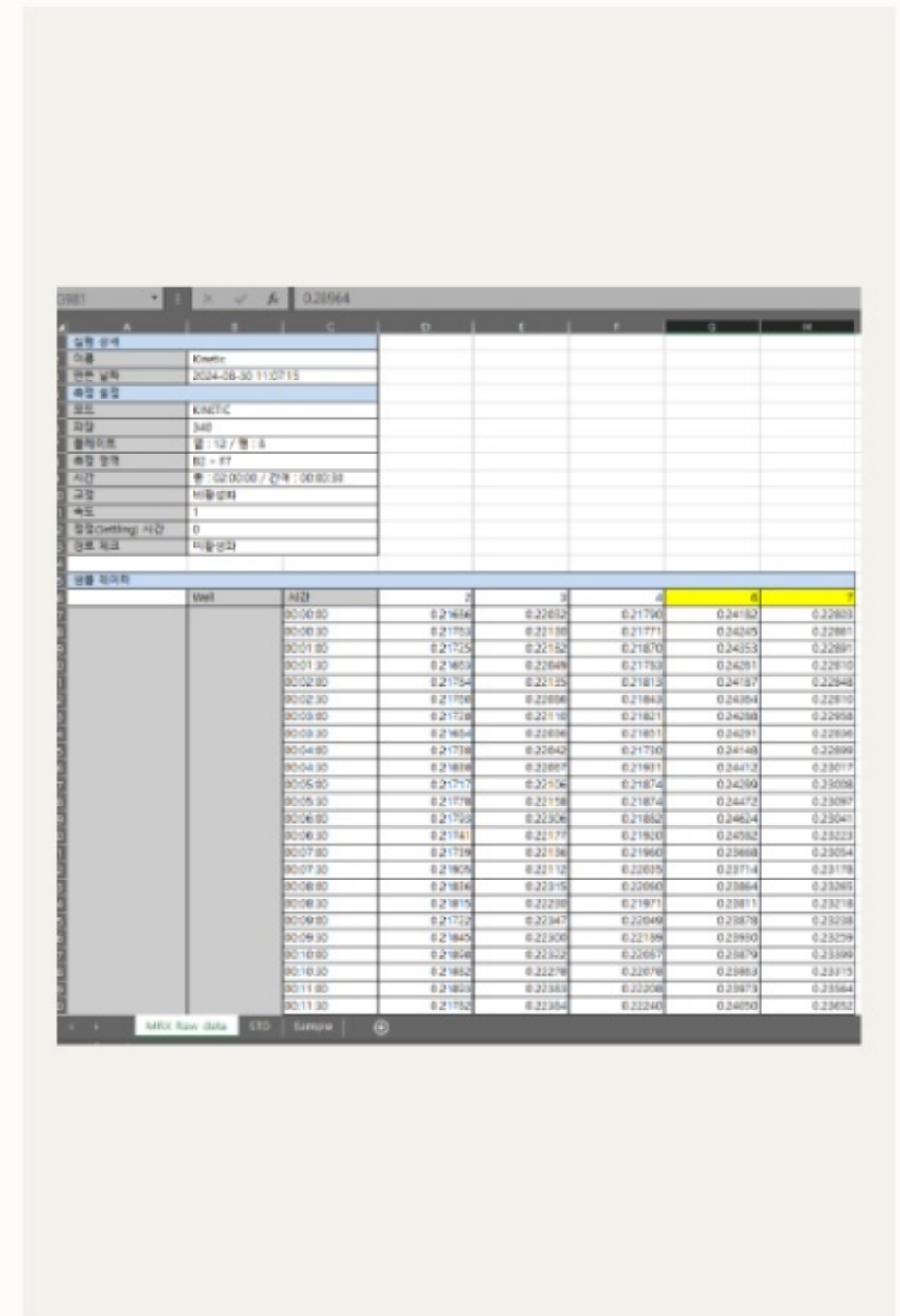
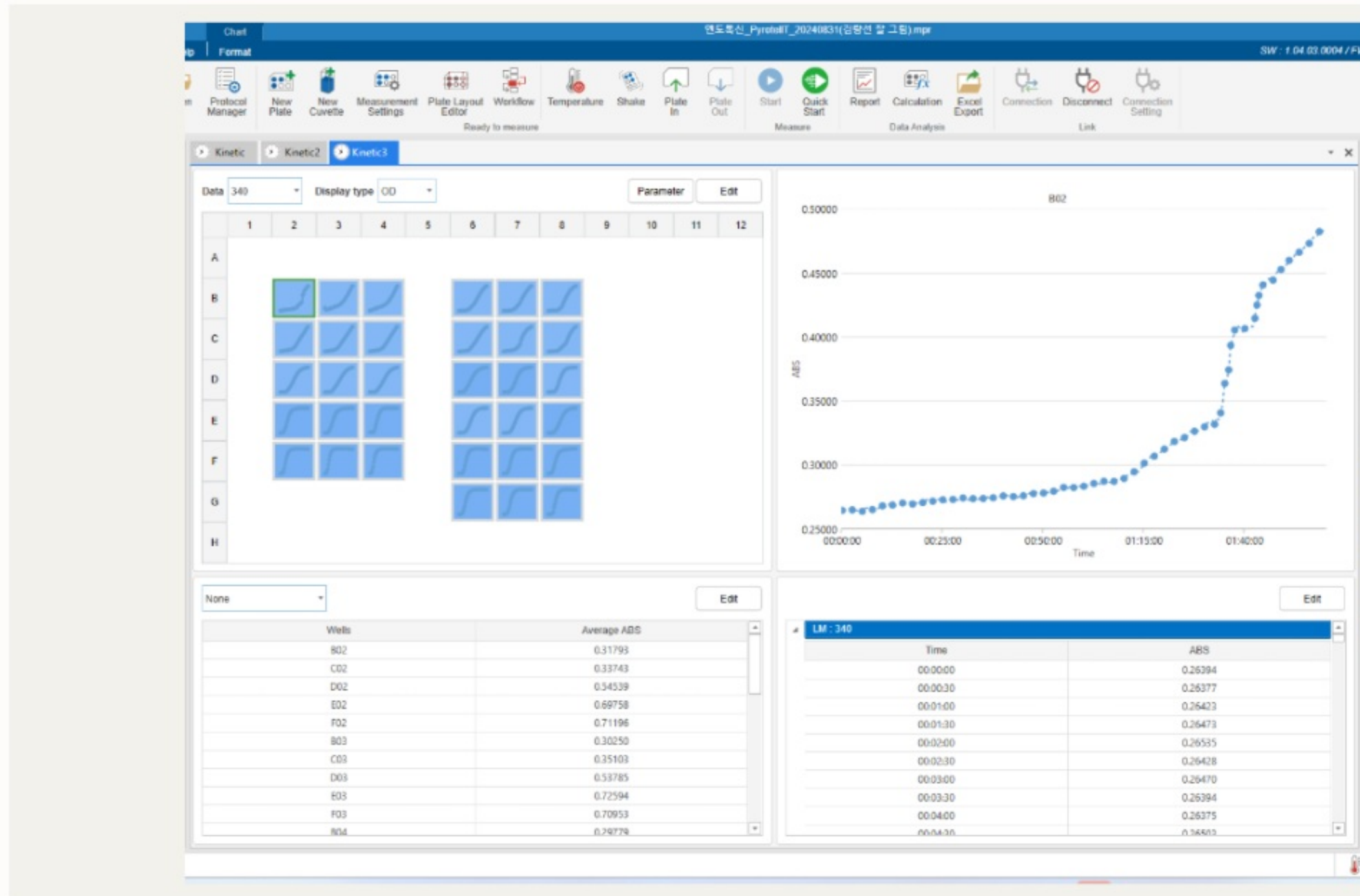
MRX New Software

Sharing K LAB's plans for the development of endotoxin analysis software

L I M I T O F K I N E T I C M O D E

Limit of Conventional Methods

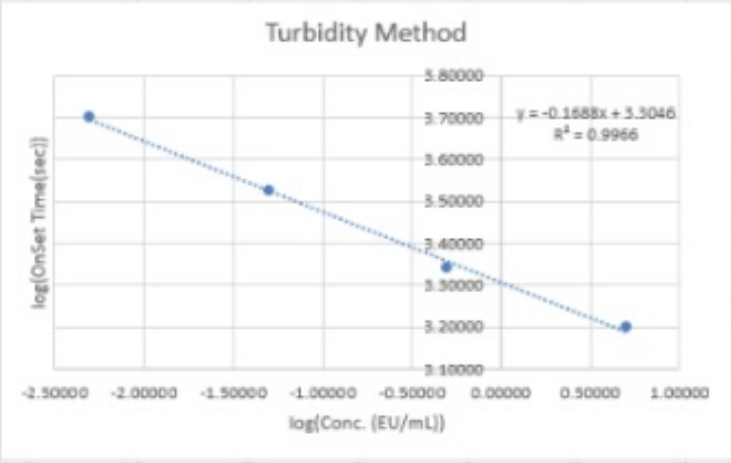
- ① External data extraction is required for comprehensive analysis.



L I M I T O F K I N E T I C M O D E

External Data Processing

G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
0.005	0.05	0.05	0.05	0.5	0.5	0.5	5	5	5		① OnSet Time										
0.21673	0.22070	0.21787	0.22170	0.23235	0.22173	0.22621	0.22428	0.22299	0.23805			NC1	NC1	NC3	0.005	0.005	0.005	0.05	0.05	0.05	0.5
0.21665	0.21966	0.21776	0.22268	0.23370	0.22304	0.22662	0.22666	0.22305	0.23925		OD Zero	0.21734	0.22124	0.21809	0.22435	0.22247	0.21676	0.22062	0.21803	0.22266	0.2326
0.21787	0.22151	0.21821	0.22294	0.23291	0.22220	0.22718	0.22724	0.22417	0.23991		Pipetting delay	600	600	600	600	600	600	600	600	600	600
0.21577	0.21982	0.21809	0.22136	0.23217	0.22195	0.22627	0.22574	0.22277	0.23797		Input values OD Threshold	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.21673	0.22147	0.21806	0.22364	0.23275	0.22314	0.22682	0.22592	0.22393	0.23930		OD Threshold	0.31733	0.32124	0.31809	0.32434	0.32247	0.31675	0.32061	0.31803	0.32265	0.3328
0.21583	0.22107	0.21910	0.22321	0.23342	0.22204	0.22755	0.22481	0.22375	0.23839		OnSet Time	6990	6990	6540	5070	5070	4890	3270	3360	3360	2130
0.21700	0.22132	0.21895	0.22283	0.23335	0.22211	0.22626	0.22612	0.22389	0.23837		Conc.(EU/mL)	-	-	-	0.005	0.005	0.005	0.05	0.05	0.05	0.5
0.21638	0.22024	0.21754	0.22273	0.23260	0.22143	0.22647	0.22409	0.22306	0.23729		log(Conc.(EU/mL))	-	-	-	-2.30103	-2.30103	-2.30103	-1.30103	-1.30103	-1.30103	-0.301
0.21597	0.22106	0.21863	0.22289	0.23301	0.22178	0.22647	0.22476	0.22413	0.23800		log(OnSet Time)	3.84448	3.84448	3.81558	3.70501	3.70501	3.68931	3.51455	3.52634	3.52634	3.328
0.21793	0.22104	0.21760	0.22323	0.23247	0.22251	0.22677	0.22557	0.22429	0.23880												
0.21640	0.22009	0.21859	0.22213	0.23141	0.22127	0.22584	0.22630	0.22518	0.23893												
0.21788	0.22174	0.21813	0.22369	0.23316	0.22322	0.22774	0.22749	0.22592	0.24010		② Standard Curve										
0.21833	0.22196	0.21894	0.22436	0.23362	0.22218	0.22684	0.22725	0.22541	0.23964		Conc.(EU/mL)	log(Conc.(EU/mL))	Avg.log(OnSet time)								
0.21793	0.22242	0.21884	0.22376	0.23391	0.22313	0.22728	0.22766	0.22500	0.23973		0.005	-2.30103	3.69977								
0.21801	0.22257	0.21921	0.22445	0.23436	0.22293	0.22738	0.22759	0.22658	0.24015		0.05	-1.30103	3.52241								
0.21805	0.22217	0.21963	0.22347	0.23592	0.22390	0.22831	0.23013	0.22686	0.24090		0.5	-0.30103	3.33834								
0.21922	0.22315	0.21951	0.22561	0.23497	0.22353	0.22865	0.23047	0.22783	0.24150		5	0.69897	3.19859								
0.21815	0.22264	0.21950	0.22509	0.23476	0.22293	0.22844	0.22943	0.22774	0.24215												
0.21882	0.22295	0.21967	0.22408	0.23520	0.22470	0.22847	0.23062	0.22798	0.24192												
0.21962	0.22225	0.22024	0.22533	0.23613	0.22455	0.22830	0.23075	0.22922	0.24302												
0.21938	0.22273	0.21962	0.22606	0.23582	0.22334	0.22962	0.23183	0.22877	0.24391												
0.21895	0.22331	0.21930	0.22590	0.23580	0.22352	0.22909	0.23252	0.23044	0.24457												
0.21998	0.22221	0.21996	0.22684	0.23876	0.22567	0.23090	0.23516	0.23373	0.24806												
0.21997	0.22343	0.21955	0.22633	0.23745	0.22506	0.23136	0.23682	0.23560	0.25201												
0.21942	0.22260	0.22023	0.22675	0.23738	0.22585	0.23037	0.23882	0.23941	0.25651												
0.21980	0.22447	0.22044	0.22635	0.23768	0.22441	0.23138	0.24221	0.24796	0.26535												
0.21941	0.22309	0.21992	0.22755	0.23895	0.22489	0.23067	0.24662	0.25747	0.27492												
0.22116	0.22460	0.22116	0.22771	0.23892	0.22696	0.23170	0.25469	0.26903	0.28747												
0.22044	0.22384	0.22056	0.22599	0.23837	0.22587	0.23163	0.26306	0.27946	0.29863												
0.22139	0.22405	0.22097	0.22680	0.23863	0.22659	0.23301	0.27214	0.29216	0.30865												
0.22267	0.22648	0.22285	0.22936	0.24103	0.22822	0.23378	0.28541	0.30480	0.32241												
0.21980	0.22464	0.22197	0.22696	0.24074	0.22714	0.23252	0.29290	0.31441	0.33284												
0.21994	0.22445	0.22137	0.22784	0.24020	0.22716	0.23236	0.30306	0.32552	0.34439												
0.22135	0.22535	0.22172	0.22871	0.24231	0.22821	0.23505	0.31449	0.33795	0.36009												
0.22175	0.22580	0.22315	0.22925	0.24422	0.22961	0.23631	0.32643	0.35113	0.37476												
0.22180	0.22555	0.22162	0.22922	0.24419	0.22858	0.23700	0.33720	0.36216	0.38855												
0.22330	0.22858	0.22474	0.23133	0.24792	0.23188	0.23893	0.35458	0.37730	0.40664												
0.22203	0.22706	0.22190	0.22834	0.24823	0.23004	0.23790	0.36557	0.38843	0.41975												



L I M I T O F K I N E T I C M O D E

“The Kinetic mode outputs only OD values over time, making it difficult to directly determine endotoxin concentration.”

0 1

Additional Calculation

In-depth analysis requires extra calculations in Excel for accurate concentration results.

0 2

Time-Consuming

The process takes extra time for data processing and interpretation.

0 3

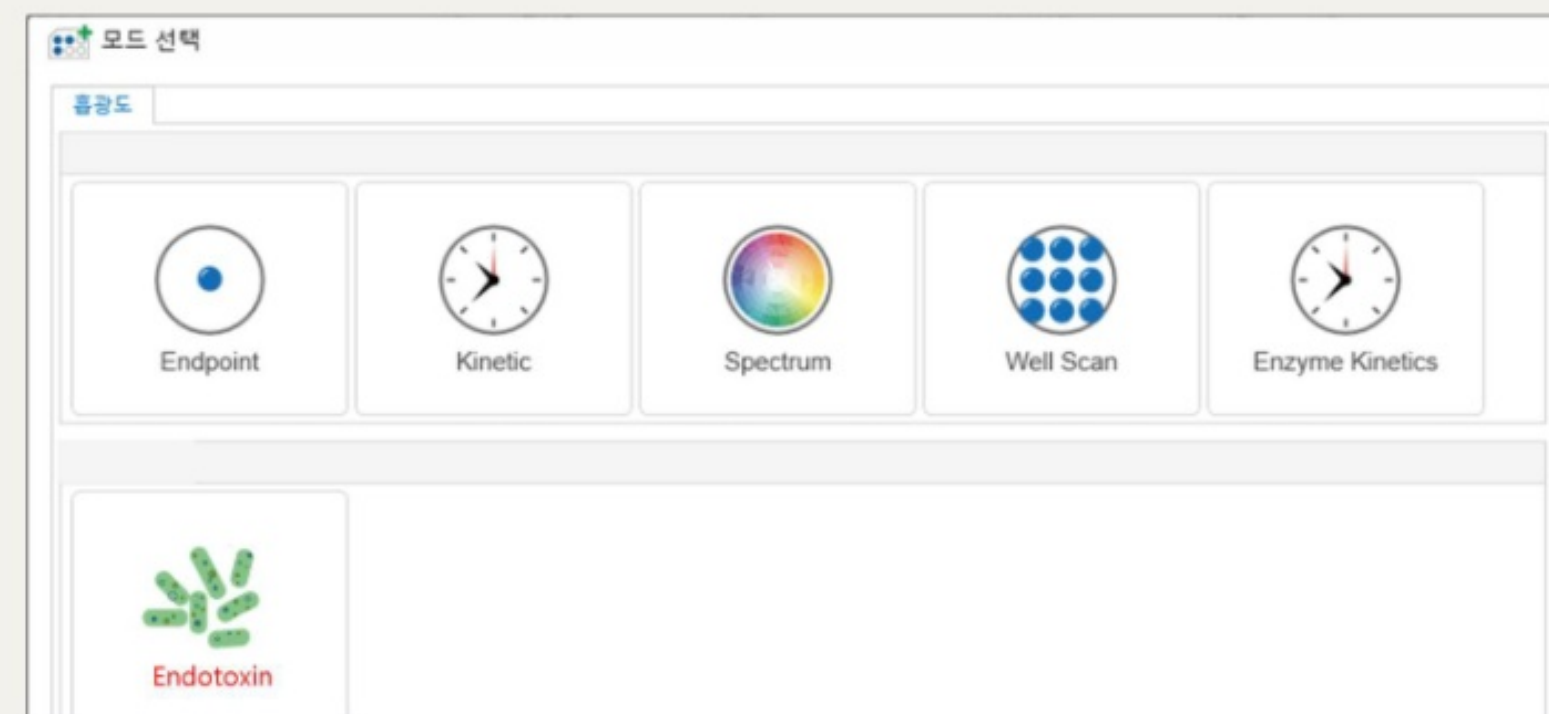
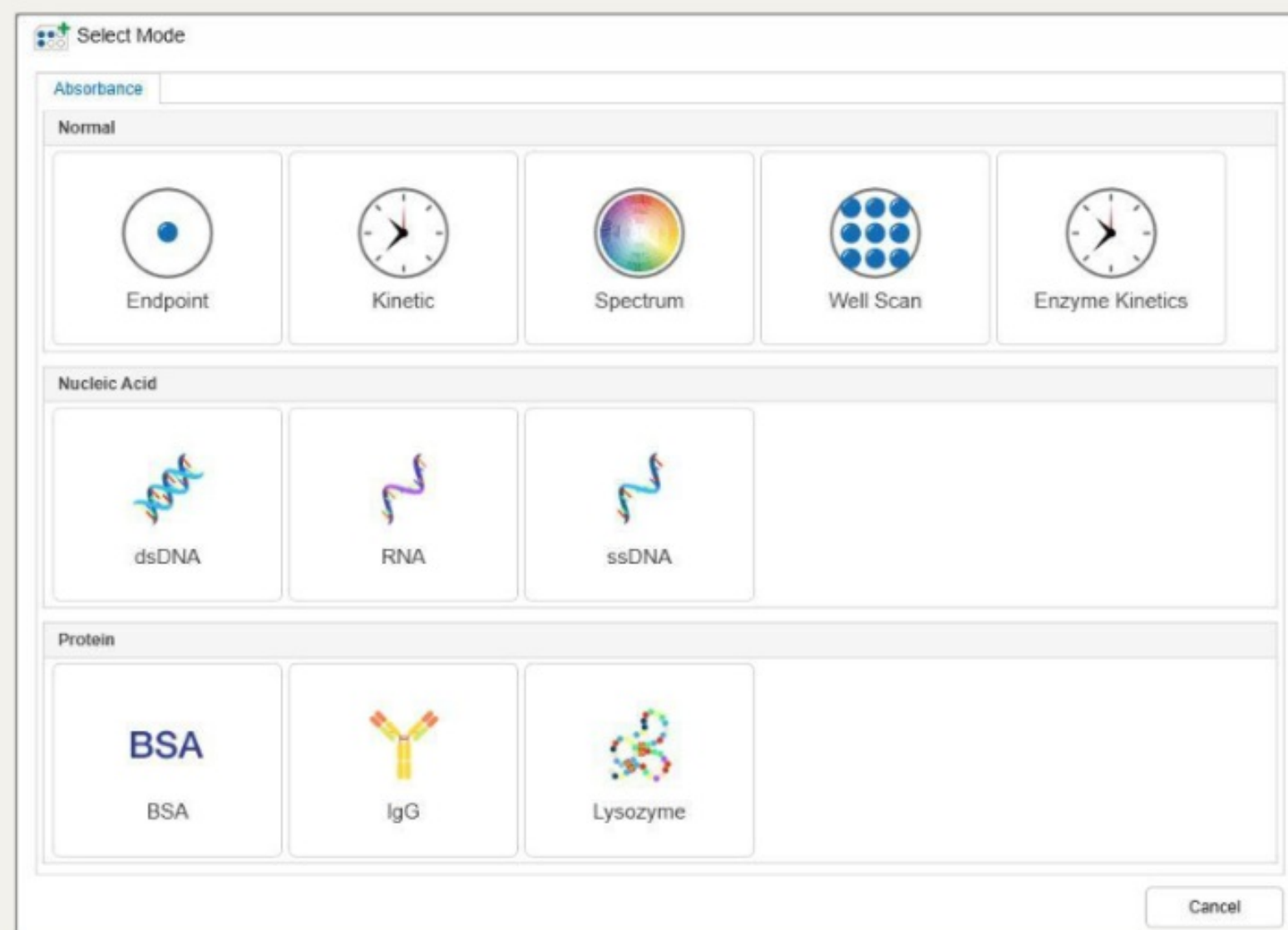
Data Integrity Issue

Complexity makes it challenging to maintain consistent and reliable data.

NEW SOFTWARE

New Endotoxin Analysis Software in MRX

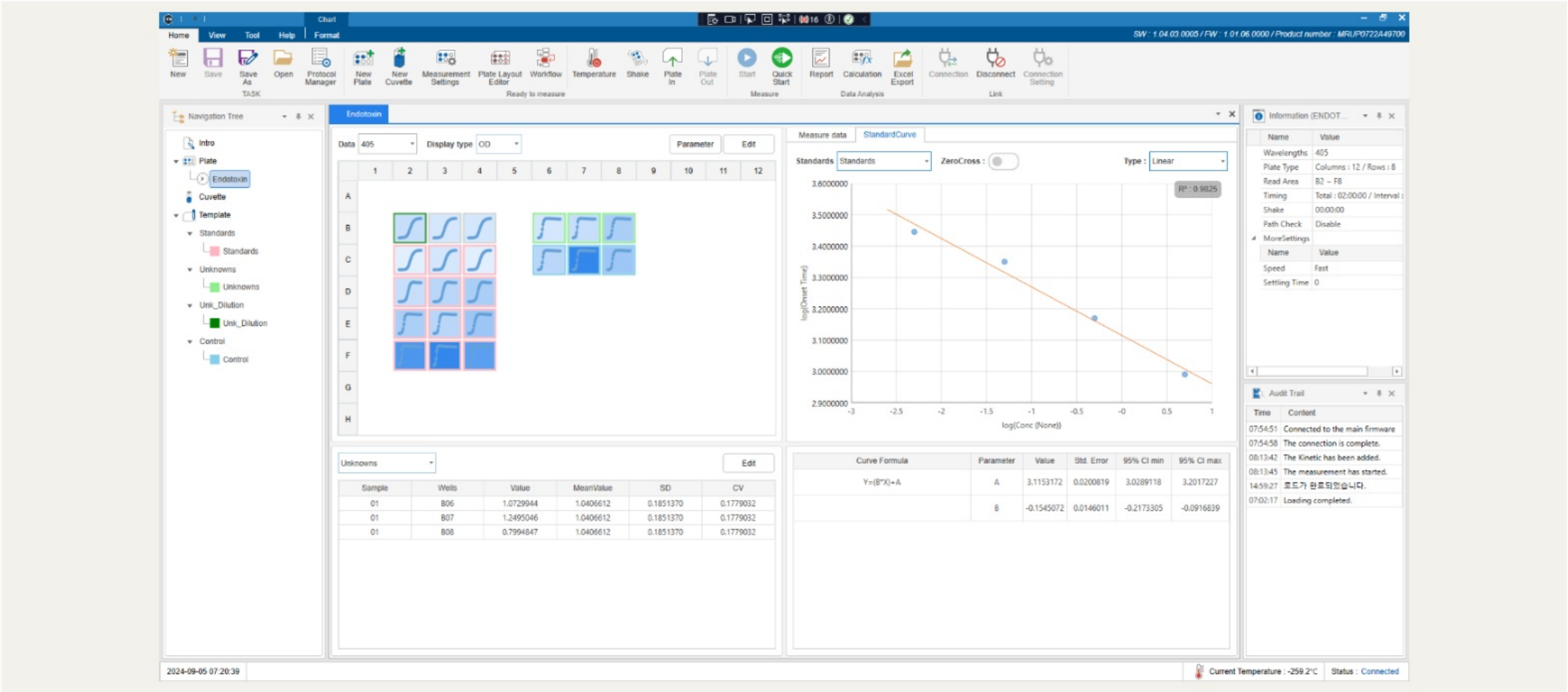
Endotoxin Application



FEATURE 01

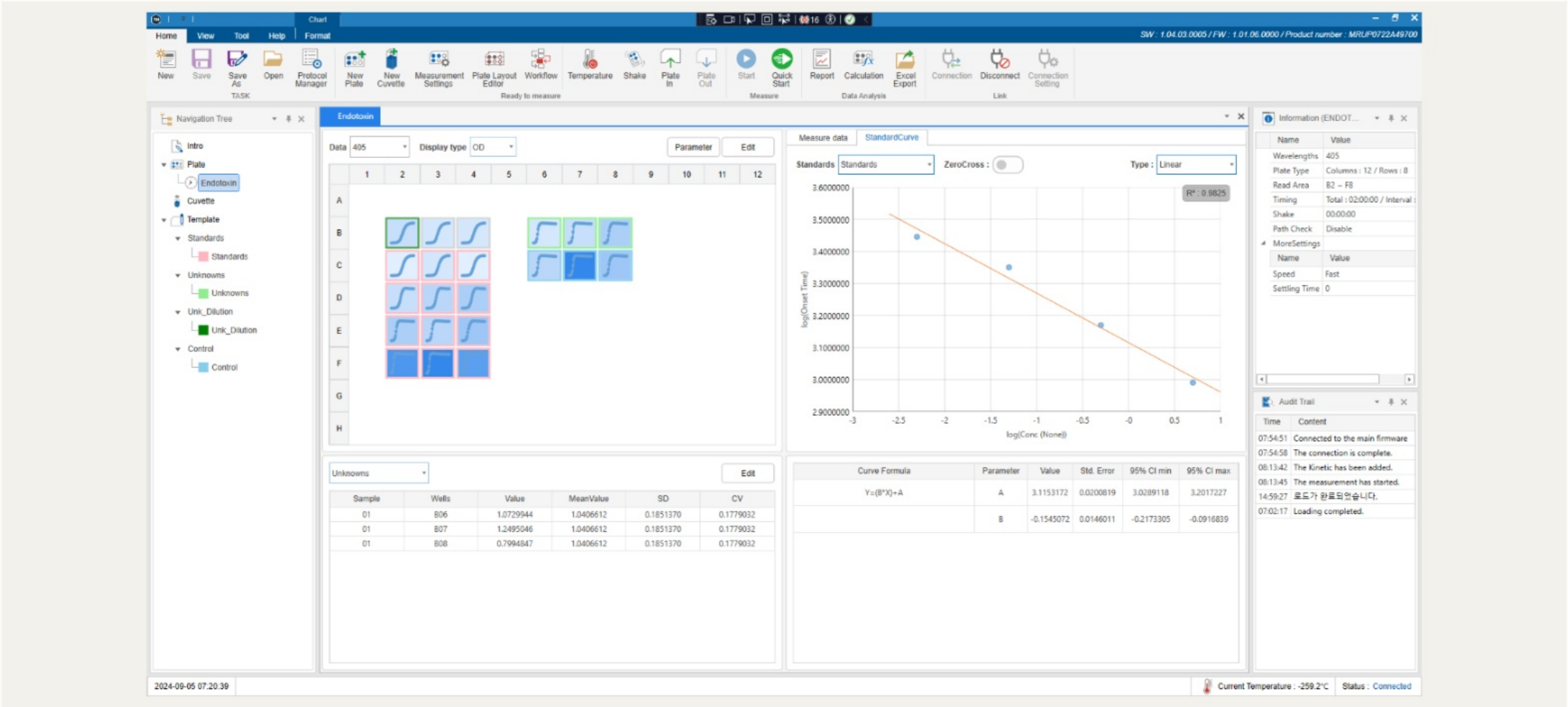
Real-Time Creation of Standard Curve

The software automatically generates the standard curve in real time during measurement, and RSQ can be checked instantly.



Automatic Calculation of Sample Concentration

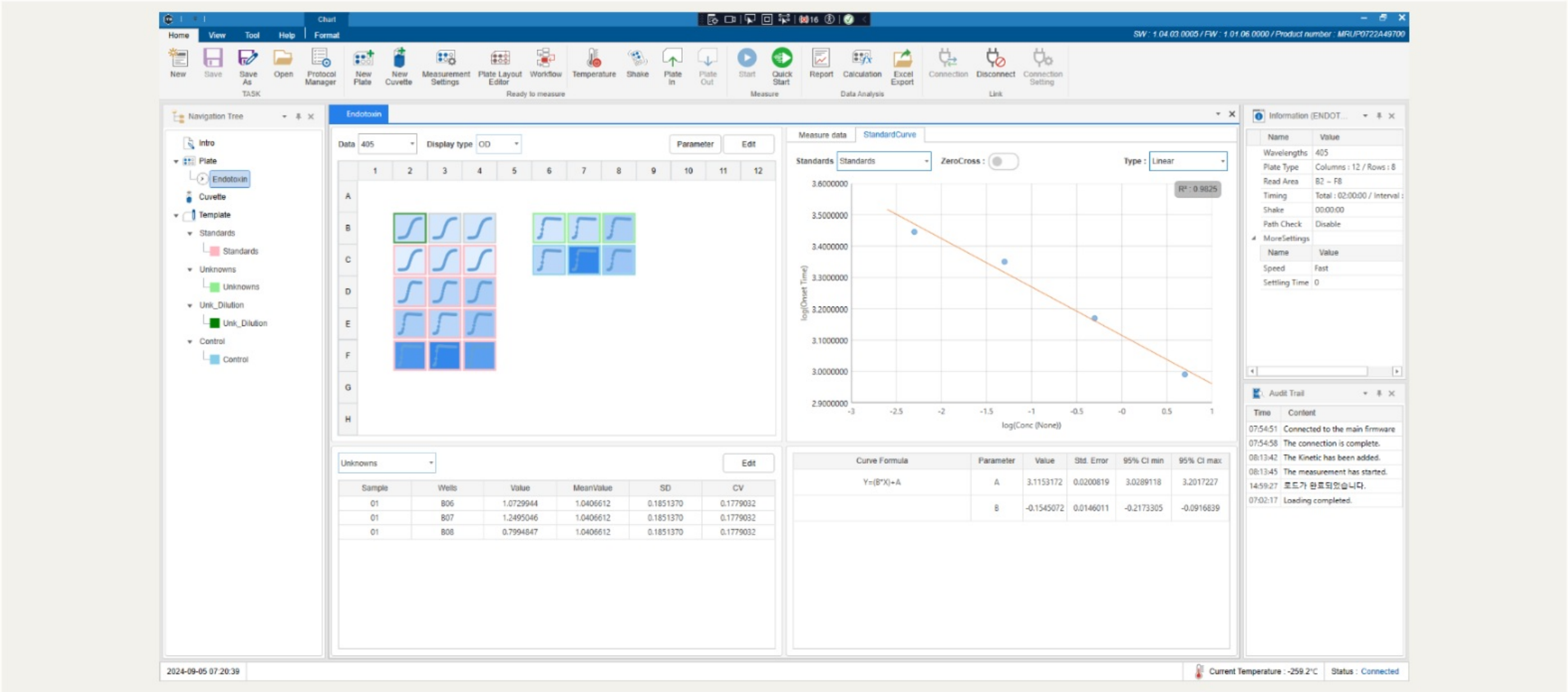
Calculates the concentration of each sample in real time.



FEATURE 03

Automatic σ and %CV Analysis

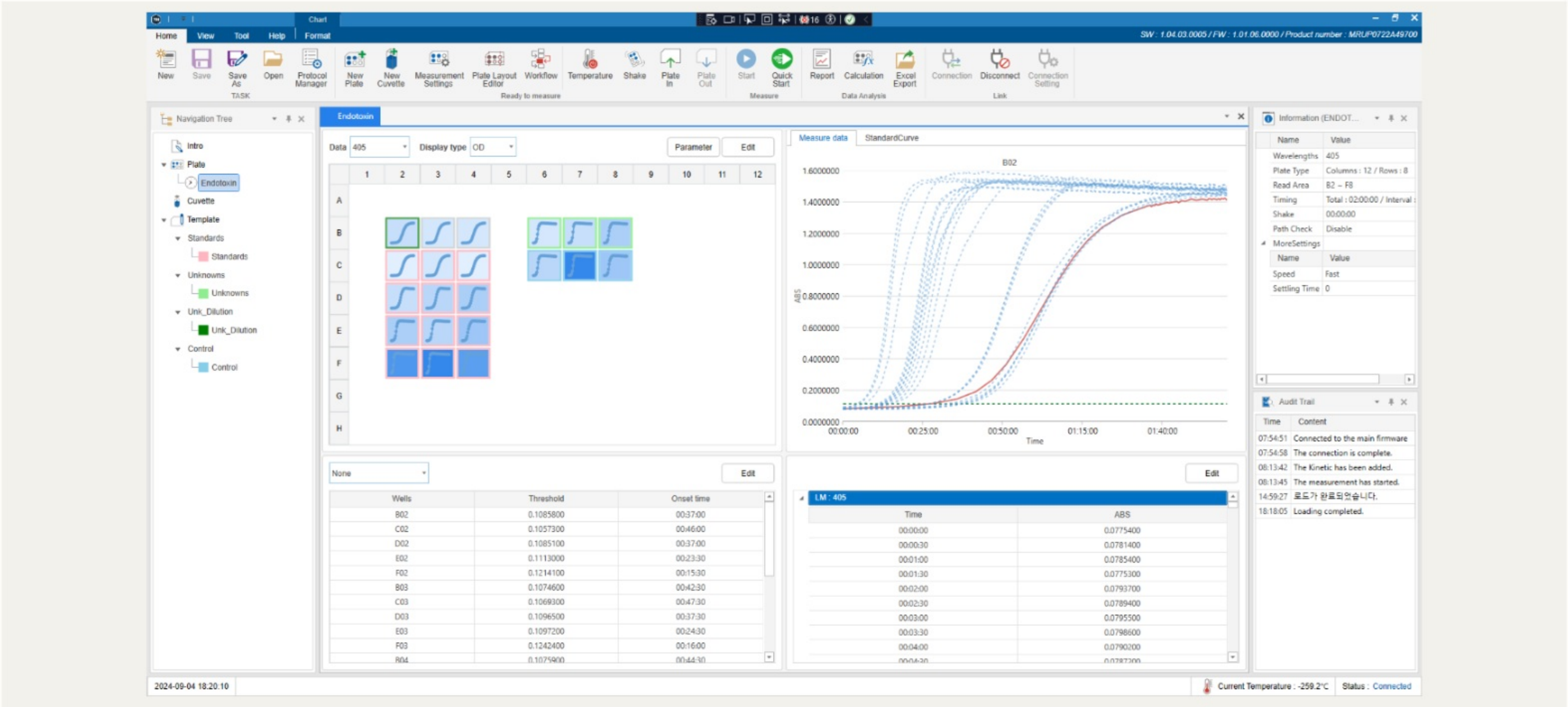
Automatically analyses standard deviation and coefficient of variation for each sample in real time, ensuring accuracy and reliability.



FEATURE 04

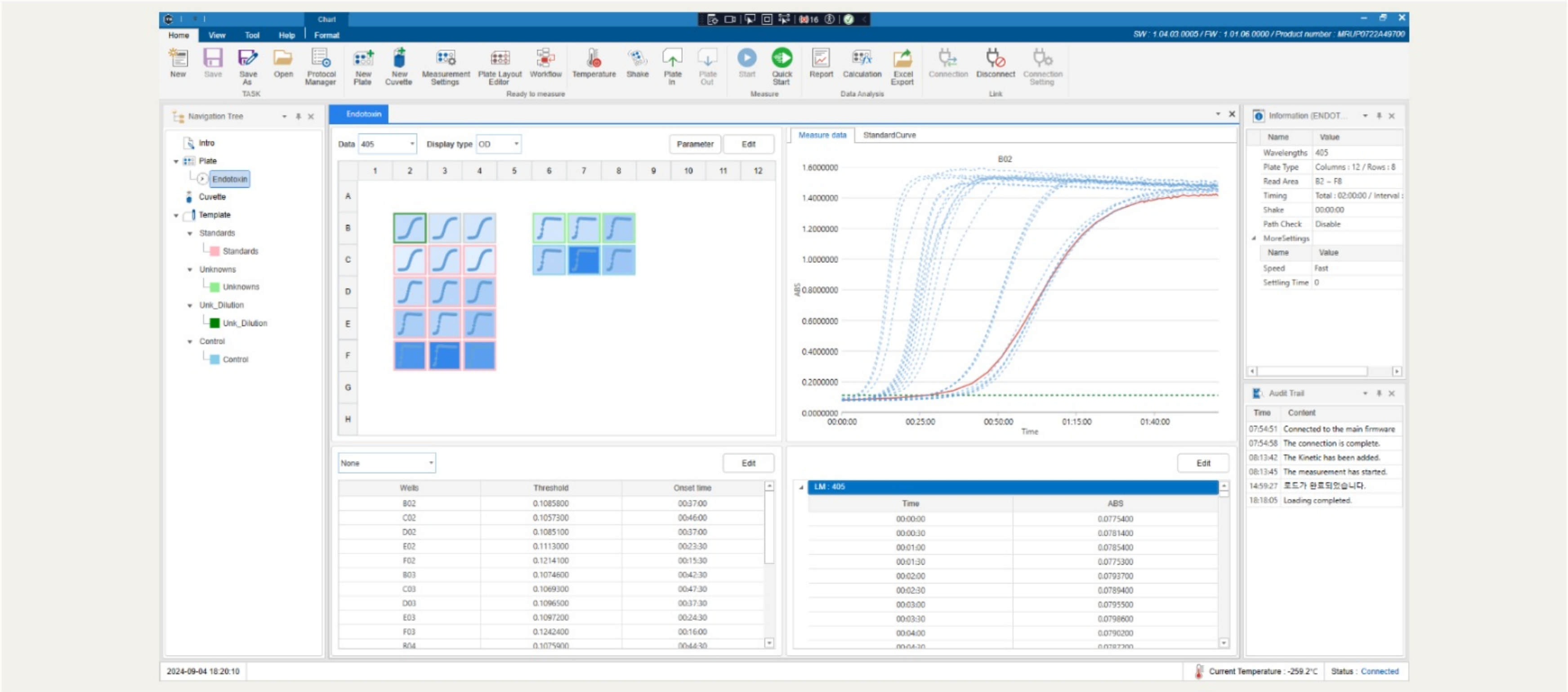
Optical Density Spectrum per Well

Real-time spectrum analysis for each well provides precise comparison of variations between samples.



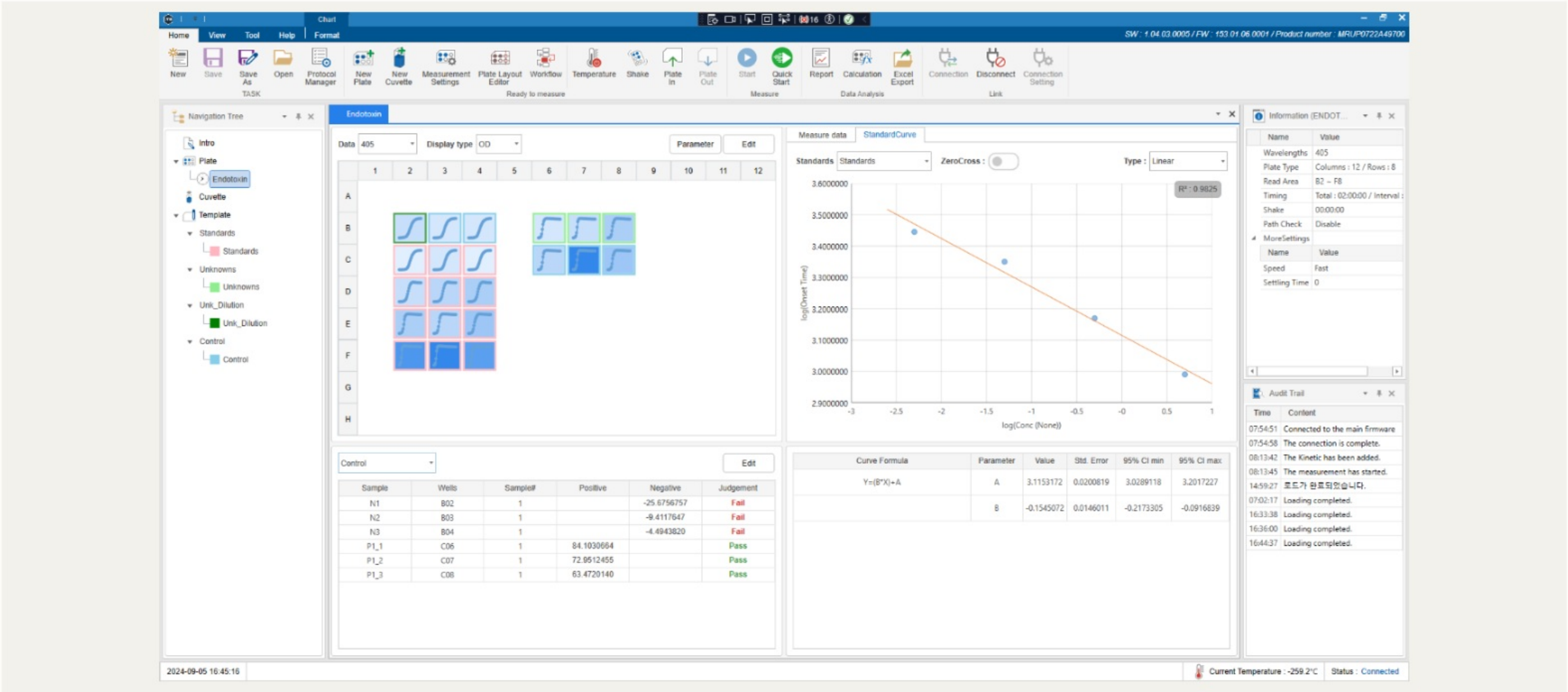
Real-Time Monitoring of All Samples

Real-time monitoring of the optical density spectrum for all samples allows rapid and accurate trend analysis.



Automatic Quality Control Analysis

PPC recovery rate and negative control separation are calculated automatically, with instant PASS / FAIL determination against user-defined references.



What the New Software Delivers

0 1

Time Efficiency

Automated calculations drastically reduce analysis time and eliminate manual processing.

0 2

Error Minimisation

Automated calculations reduce human errors associated with manual data entry.

0 3

Standardised Consistency

Unified protocols ensure consistent, standardised results across labs.

0 4

Real-Time Processing

Live data provides immediate feedback, aiding quicker decision-making.

G L O B A L S T R A T E G Y

K LAB's Global Strategy

Expanding Market Reach

K LAB is committed to expanding its presence in key markets, including Asia, Europe, and the Americas.

Tailored Solutions

Customised solutions address the specific needs of each region, ensuring optimal performance and satisfaction.



S U P P O R T & S E R V I C E

Customer Support and Service

Technical Support

Our experts offer quick, accurate solutions for endotoxin analysis and spectrophotometry issues.

Software Updates

Regular updates ensure access to the latest features and security patches.

Installation & Maintenance

Tailored installation, setup, and regular maintenance to optimise lab equipment performance.



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